

WASP-52b

R-1510

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_191008 · created 2026-08-01T15:53:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458764.8078 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1663 +/- 0.0098
Transit depth (Rp/Rs)^2	2.77 +/- 0.33 [%]
Orbital Inclination (inc)	86.55 +/- 1.17
Airmass coefficient 1 (a1)	1.012 +/- 0.011
Airmass coefficient 2 (a2)	-0.0069 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#1 - [395, 166]
Optimal Method	PSF photometry
Transit Duration (day)	0.0809 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1663 ± 0.0098 vs 0.1646 (deviation 0.17σ)
Duration fit vs published	0.0809 d vs 0.0758 d (deviation 1.01σ)
Mid-time O–C	-3.3 min
Depth SNR	17.0
Residual scatter	1.11 %

Light curve

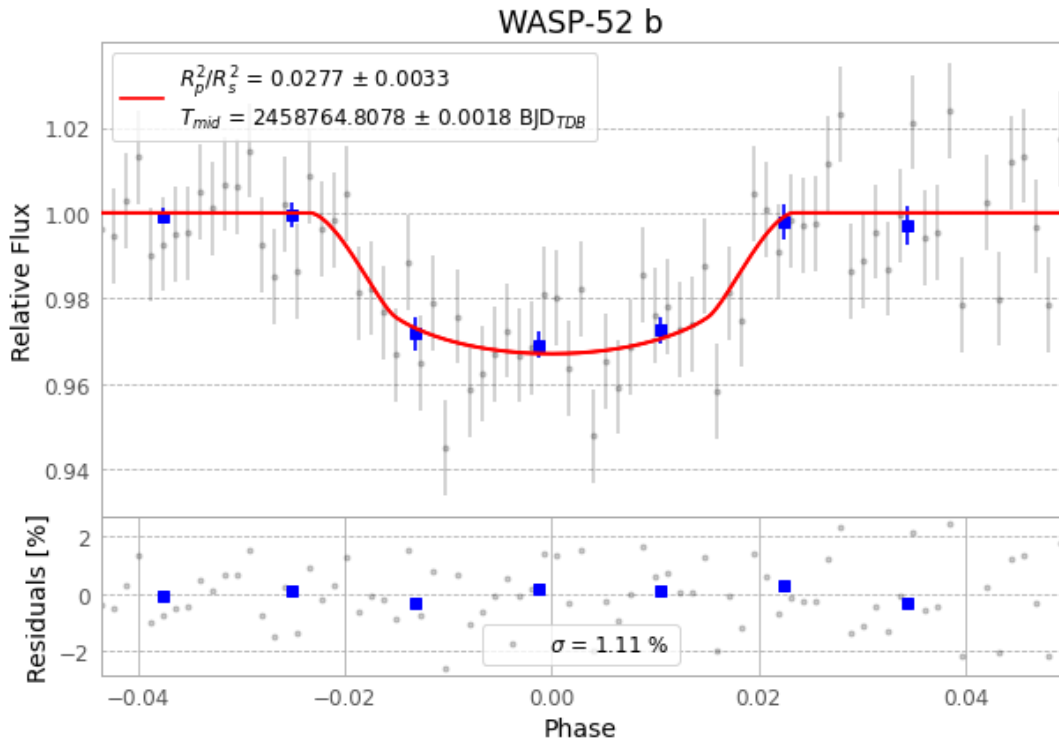


Figure 1 — FinalLightCurve_WASP-52 b_2019-10-07.png (SHA-256 7590a30b99c1a3ef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [211, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 392a7368882e9616...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), WASP-52191008052412.FITS → WASP-52191008091813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-191008.log (5dc53fcdbee4...), FinalLightCurve_WASP-52 b_2019-10-07.png (7590a30b99c1...), FinalLightCurve_WASP-52 b_2019-10-07.csv (832263ea3b28...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:53Z.